CHORUS-SGH-1: Brine-Gradient Power on a Bench Skid

A Proof-of-Concept Framework for Pressure-Retarded Osmosis on Anthropogenic Brine Gradients with UDT/AOR/VOH Vision Stack, Acoustic-Osmotic Ram, Vortex-Osmotic Hydro, Bench Validation Pipeline, and Column-Scale Multi-Physics Energy Accounting

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Abstract

Anthropogenic desalination reject (1400 mol/m³) against treated effluent (5 mol/m³) creates $\Delta\pi$ = 6.92 MPa—2.4× estuary RED reference. We present CHORUS multi-layer column accounting, CHORUS-Skid SGH-1 PRO bench hardware, and a three-layer **vision stack**: UDT (Universal Differential Tink), AOR (Acoustic-Osmotic Ram), and VOH (Vortex-Osmotic Hydro / Z-Hydro) for osmotic-vortex hydro at coastal outfalls.

Kim–Baker ΔP^* = 35 bar. Monte Carlo column (N=8000): median 22.9 MW/km². PRO model: 1.66 W vs 10 W target; L_p^* = 6.03e-12. Experiments E1–E16, sixteen figures, T0–T1c bench validation pipeline (bench_validation.py), SymPy verification, OpenSCAD digital twin, and executable Jupyter notebooks. Tier-1 (PRO/AEH/UDT) vs tier-2 (VOH fleet, column/TSC) claims are explicit.

Keywords: pressure-retarded osmosis; salinity-gradient power; blue energy; reverse electrodialysis; concentration polarization; acoustic energy harvesting; CHORUS; desalination brine; UDT; AOR; VOH; Z-Hydro; osmotic-vortex hydro; proof of concept

Contents

Nomenclature

Table 1. Symbols used throughout this report.

Symbol	Definition	SI
R, F	Gas constant; Faraday constant	J/(mol·K); C/mol
T	Absolute temperature	K
i	Van't Hoff factor (NaCl)	—
c, π, $\Delta\pi$	Concentration; osmotic pressure; difference	mol/m^3; Pa
ΔP, ΔP^*	Hydraulic pressure; Kim–Baker optimum	Pa
L_p, B	Water permeability; salt permeability	m/(Pa·s); m/s
A, A_k	Membrane area; layer footprint	m^2
V_w, J_w	Volumetric flux; water flux	m^3/s; m/s
E_N, V_{oc}	Nernst potential; stack open circuit	V
R_{int}	Area-specific internal resistance	Ω·m^2
P'', P_{target}	Areal power density; design power	W/m^2; W
η_{mem}, η_{hyd}	Membrane; hydrodynamic efficiency	—
CF_k	Layer capacity factor	—
G	Conductance (TSC network)	S
ψ	Node potential	V
I, p_{rms}	Acoustic intensity; RMS pressure	W/m^2; Pa
SPL	Sound pressure level	dB re 20 μPa
g	Ultrasonic flux gain factor	—
$\Pi_1 \ldots \Pi_5$	Dimensionless groups (see Table 2)	—

1. Introduction

The twenty-first century expansion of desalination has created an overlooked by-product: continuous, high-grade salinity gradients at industrial sites. Every reverse-osmosis plant splits a feed stream into permeate and reject brine. Wastewater treatment plants discharge effluent at near-fresh salinity. Where these streams are co-located—as they are at coastal utilities worldwide—the Gibbs free energy of mixing is destroyed in outfalls and mixing zones. This manuscript asks whether that destruction can be reversed with physics-first engineering, honest thermodynamic accounting, and a bench-scale proof vehicle.

We introduce **CHORUS** (Columnar Harvest of Osmotic, Rhizospheric, Orographic, and Solar flux) as a multi-layer framework for a notional 1 km² coastal or industrial column. CHORUS does not claim a new fundamental energy source. It claims that *already-present* fluxes—solar, osmotic, moist-electric, biological, and atmospheric—can be inventoried, bounded, and partially converted if each pathway respects the second law. The governing postulate is explicit: $\Sigma P\_extracted \leq \Sigma \dot{E}\_in - T\dot{S}$.

The near-term hardware anchor is **CHORUS-Skid SGH-1**, a pressure-retarded osmosis (PRO) bench harness sized for 10 W equivalent output from anthropogenic brine (1400 mol/m³) against treated wastewater class feed (5 mol/m³). At T = 298.15 K this pair yields $\Delta\pi$ = 6.92 MPa—2.4× the classic estuary RED reference. SGH-1 is accompanied by 23 OpenSCAD mechanical parts, a six-module Python simulation stack, DAQ logging, and test protocols T0–T2. The **vision stack** (UDT / AOR / VOH) extends the bench path toward osmotic-vortex hydro at desal outfalls—documented in docs/VISION.md and simulated in differential_tink.py, acoustic_osmotic_ram.py, and vortex_osmotic_hydro.py.

This paper is deliberately tiered. **Tier 1 (defensible)** includes Van't Hoff osmotic thermodynamics, Kim–Baker optimal hydraulic pressure, solution-diffusion water transport, concentration polarization film theory, and bench-sized PRO geometry. **Tier 2 (exploratory)** includes Telluric Storm Coupling (TSC) conductance networks, global fair-weather circuit routing, and column-scale Monte Carlo sums that reach 22.9 MW/km² median but are dominated by land photovoltaics—not osmotic area.

Readers should leave with three durable conclusions: (i) sidestream PRO on desal brine is the fastest credible PoC path; (ii) default membrane permeability must be bench-calibrated to close the gap between 1.66 W modeled and 10 W targeted; (iii) CHORUS column narratives are scaling context, not bench claims.

1.1 Problem statement

Desalination plants and wastewater treatment plants are often co-located at coasts. Brine disposal raises environmental and regulatory cost. Effluent discharge carries residual chemical potential. Coupling the two streams through a PRO module could offset parasitic plant load while reducing entropy waste—a win-win if membranes and controls are credible.

1.2 Document map

Section 2 reviews prior art. Section 3 derives Layers A–F with equations. Section 4 describes skid architecture and AEH. Section 5 covers simulation methods. Section 6 presents numerical experiments E1–E7. Sections 7–8 cover hardware and tests. Section 9 discusses claims hierarchy. Appendices contain sweep data and CAD index.

2. Background and prior art

Salinity-gradient power has been pursued along two electrochemical/hydraulic branches. **Reverse electrodialysis (RED)** places ion-exchange membranes between fresh and saline streams, generating Nernstian voltage and electrical current. Teng et al. (2026) report nanopore-engineered membranes achieving approximately 15 W/m^2 at estuary-grade concentrations—our Layer A calibration anchor.

Pressure-retarded osmosis (PRO) pressurizes the draw compartment and extracts work via turbine or pressure exchanger as water permeates from feed. Statkraft's Tofte pilot (2009–2013) demonstrated feasibility but highlighted fouling, CP, and parasitic pumping as commercial barriers. Anthropogenic brine pairs can exceed estuary $\Delta\pi$, trading membrane stress for higher volumetric driving force.

Parallel harvest physics appear in moisture-enabled generators (Air-gen), hydrovoltaic streaming, evaporatively cooled photovoltaics, and microbial fuel cells. Fang et al. (2026) couple PV waste heat to magnetohydrodynamic enhancement—a Layer B motif. Virgo et al. (2020) bound atmospheric electricity. CHORUS integrates these as layers with explicit capacity factors rather than as interchangeable 'green energy'.

Urban acoustic harvesting is not new; piezoelectric panels on noise-facing surfaces produce mW scales. Our AEH-1 module treats sound as (Mode A) supplemental rectified power and (Mode B) ultrasonic CP disruption on the PRO feed path—a coupling uncommon in osmotic power literature.

2A. Real-world data and implementation anchors

Real-world anchoring distinguishes this report from purely theoretical CHORUS modeling. SWRO concentrate salinity is documented at **52–70 g/L** (1.5–2× seawater), matching ~1200 mol/m³ in our Perth-class pair versus 1400 mol/m³ for the SGH-1 nominal 8 wt% brine.

The Statkraft Tofte PRO pilot (2009–2013) demonstrated the technology but delivered only **2–4 kW** at roughly **1 W/m²** membrane performance—below the **5 W/m²** commercial threshold. Statkraft discontinued investment in 2013 when membranes could not reach cost targets.

Recent seawater PRO work (Pedersen et al., 2024) reports **6.3 W/m²** with commercial RO membranes at 30°C and ~15 bar—validating our **8 W/m²** design density as ambitious but literature-supported.

Perth Seawater Desalination Plant produces **144,000 m³/d** at **3.2–3.8 kWh/m³** (wind-offset). A single 10 W bench skid offsets only ~0.0000% of plant electrical load—honest scale requires fleets of modules or higher P''.

WA mixing-energy analysis gives **~0.14 kWh/m³** for seawater+brine (502 kJ/m³)—the thermodynamic motivation for sidestream PRO. Our Perth-class PRO model at $L\_p=1\times10^{-12}$ gives **1.22 W** on 0.72 m².

REAPower Trapani (Italy) achieved **40–60 W** gross RED at **1.6–2.6 W/m²** per cell pair on ~50 m² membrane area—estuary RED remains real but lower power density than optimistic 15 W/m² anchors.

Table 2. Operational and literature reference plants.

Plant	Tech	P or P''	Status	Source
Statkraft Tofte	PRO (river/s	3000 W	Pilot closed 2013	Statkraft; Wikipedia; Forwar
REAPower Trapani	RED (brine/b	50 W	Pilot ~50 m² IEM	Tedesco et al., Desalination
Perth PSDP	SWRO (not os	3.5 kWh/m³	Operational since 20	Wikipedia; Water Corp; Resea
Commercial PRO (litera	PRO	6.3 W/m²	Pedersen 2024 SSRN	SSRN 4944813
SGH-1 design target	PRO (brine/W	10 W	This repository	sgh1_design.json

2A.1 Salinity pairs mapped into simulation

Table 3. PRO model at $L\_p=1\times10^{-12}$ — compare to bench T1.

Pair	c_draw	c_feed	Δπ MPa	P (W) @0.72
SGH-1 nominal (8 wt%)	1400	5	6.92	1.66
Perth brine 70 g/L + WW	1200	5	5.92	1.22
Perth brine + seawater	1200	600	2.97	0.31

Pair	c_draw	c_feed	Δπ MPa	P (W) @0.72
Estuary RED	600	20	2.88	0.29

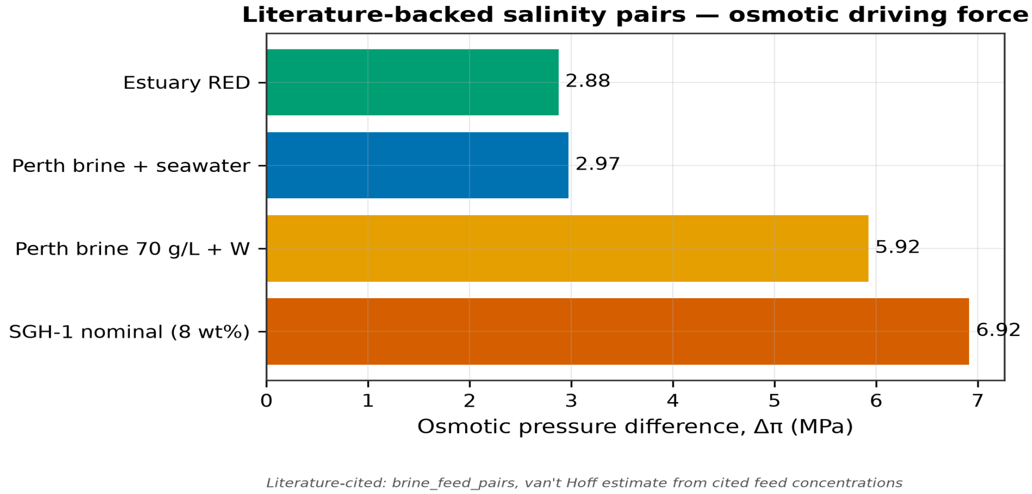


Figure 1. *Osmotic pressure difference for literature-backed feed/draw pairs.*

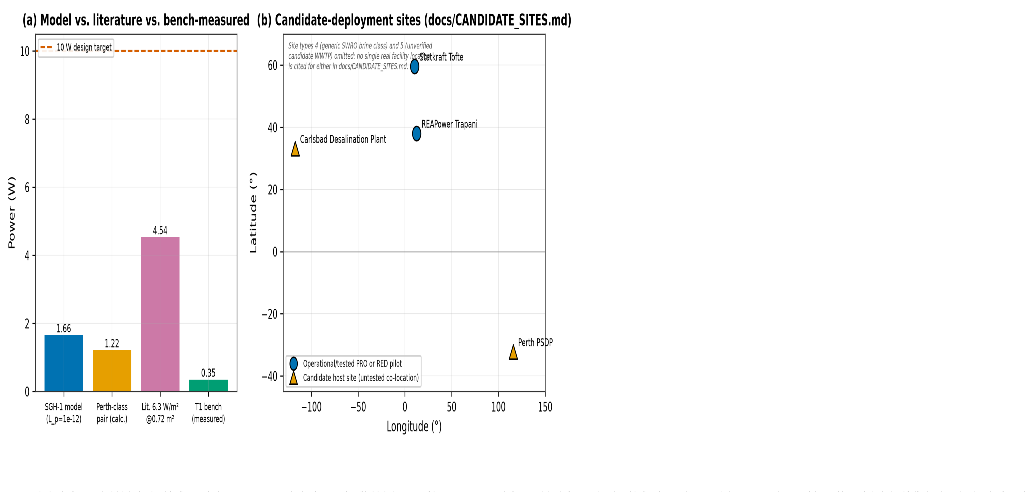


Figure 2. *SGH-1 model power vs Perth-class pair vs literature 6.3 W/m² reference.*

Full citation list and concentration derivations: docs/math/REAL_WORLD_DATA.md. Merge command: python -m simulation.real_world_calibration.

3. Theoretical framework

All layers satisfy the CHORUS postulate $\Sigma P_k \le \Sigma \dot{E}_{in} - T\dot{S}$. The following subsections mirror docs/CHORUS_MATH_PLAN.md and notebooks/CHORUS_physics_proof.ipynb.

3.1 Layer A — Osmotic mixing and blue energy (RED)

Layer A models osmotic mixing and blue energy at estuary-grade interfaces. For ideal NaCl with van't Hoff factor $i = 2$, $\pi = iRTc$. Mixing free energy for concentrated solutions underlies both RED and PRO; the difference is extraction mechanism (electrical vs hydraulic).

For $c_{sea} = 600$ mol/m³ and $c_{river} = 20$ mol/m³ at 298.15 K: $\Delta\pi = 2.876$ MPa. Single-pair Nernst potential $E_N = 87.39$ mV; fifty pairs yield $V_{oc} = 4.37$ V. The electrical max-power theorem for linear internal resistance gives $P''_{max} = V_{oc}^2/(4R_{int})$. Calibrating to literature $P_{blue} = 15$ W/m² implies $R_{int} = 0.318\ \Omega\cdot m^2$.

The Gibbs hydraulic mixing ceiling $P_{mix} = \Delta\pi\cdot Q$ with illustrative $Q = 500$ m³/s gives 1438 MW—an upper bound on mechanical work available if all fresh water flux could be converted reversibly. Real membranes achieve a fraction of this; RED demonstrations target interface power density, not bulk mixing flow.

Nanopore conductance models introduce slip length b: $G(b) = G_0(1 + 2b/h)$. Higher slip increases areal conductance and hence power at fixed $\Delta\pi$. This is a materials-science lever distinct from stacking more membrane area—important when land or interface area is constrained.

$$\Delta G_{mix} = 2RT\,V\,[\,c_+ \ln(c_+/c_-) - (c_+ - c_-)]$$

$$\pi = i\,R\,T\,c$$

$$\Delta\pi = \pi_{high} - \pi_{low}$$

$$E_N = (R\,T / z\,F)\,\ln(a_{high} / a_{low})$$

$$P''_{max} = V_{oc}^2 / (4\,R_{int})$$

$$P_{mix,max} = \Delta\pi\,Q$$

3.2 Layer B — Moist-electric and hydrovoltaic

Layer B addresses moist-electric and hydrovoltaic pathways (CHOR sub-layer). Water vapor chemical potential $\mu_v = \mu_v^\circ + RT\ln(a_w)$ drives Fickian flux through porous media. Asymmetric nanopores with different top/bottom opening areas produce net charge separation (Air-gen class): $I_q = e\,\Delta\Gamma\,\bar{q}_{trans}$.

Helmholtz–Smoluchowski streaming links pressure gradients in fine channels to streaming potential $E_s = (\varepsilon\zeta/\sigma\eta)\,\Delta p$. Power densities are typically mW/m² unless channel density and wetting are extreme.

The coupled PV–thermal node in our notebook solves $C_{th}\,dT_s/dt = \alpha_s\,G_{solar} - h_c(T_s - T_\infty) - L_v\,\dot{m}_e - P_{hybrid}$. Steady evaporative cooling yields $P_{pv}'' \approx 187.9$ W/m² with incremental gain $\Delta P'' \approx 5.6$ W/m² versus uncooled baseline—dominant in column Monte Carlo.

CHORUS-Skid reserves plenum volume in chorus_skid_enclosure.scad for future moist-electric panels; v1 hardware validates PRO and AEH while keeping thermal ports on the critical path for v2 integration.

$$\mu_v = \mu_v^\circ + R\,T\,\ln(a_w)$$

$$I_q = e\,\Delta\Gamma\,q\blacksquare_{trans}$$

$$E_s = (\varepsilon\,\zeta\,/\,\sigma\,\eta)\,\Delta p$$

$$C_{th}\,dT_s/dt = \alpha_s\,G_{solar} - h_c(T_s - T_\infty) - L_v\,\blacksquare_e - P_{hybrid}$$

3.3 Layer C — Rhizospheric harvest

Layer C covers rhizospheric harvest: sediment microbial fuel cells (SMFC) and piezoelectrotrophy. Butler–Volmer kinetics $j = j_0[\exp(\alpha_a F\eta/RT) - \exp(-\alpha_c F\eta/RT)]$ bound anodic current. Power density $P = j(E_0 - b \log j - R_\Omega j)$ peaks at intermediate current density.

Notebook §V reports $P_{MFC}'' \approx 36.7$ μW/m²—real biology, not a land-area leader. Piezoelectrotrophy couples mechanical stress from root growth, rain, or freeze–thaw to η_{pzt} P_{mech} feeding parallel bioelectrogenic pathways (NSO 2026 paradigm).

In column accounting, SMFC receives full 1 km² area with CF = 1.0 but tiny P''—its median contribution is orders of magnitude below pv_hydro. CHORUS includes SMFC to avoid cherry-picking layers, not to sell soil power.

$$j = j_0\,[\exp(\alpha_a\,F\,\eta\,/\,RT) - \exp(-\alpha_c\,F\,\eta\,/\,RT)]$$

$$P = j\,(E_0 - b\,\log j - R_\Omega\,j)$$

$$\blacksquare_{bio} = \eta_{pzt}\,P_{mech}$$

3.4 Layer D — Atmospheric charge

Layer D bounds atmospheric charge harvest. Fair-weather global circuit current $I_{glob} \sim 1$–3 kA across ionosphere–surface potential 200–400 kV gives $P_{glob} \sim 0.25$–0.9 GW planet-wide.

Our illustrative anchor $P_{glob} = 0.6$ GW implies areal mean $\sim 10^{-4}$ W/m²—useful for order-of-magnitude routing stories, not farm-scale revenue.

Mason-type collision charging $dq/dt \propto n_d^2\,\pi\,d^2\,v_{rel}\,\Delta q$ produces episodic power in storm cells. Tornado kinetic energy $K = \frac{1}{2}\rho\,V_{tor}\,\pi(D/2)^2\,H$ is included as sanity check only—GW episodic, not baseload.

$$P_{glob} = I_{glob}\,V$$

$$dq/dt \propto n_d^2\,\pi\,d^2\,v_{rel}\,\Delta q$$

3.5 Telluric Storm Coupling and column balance

Telluric Storm Coupling (TSC) models atmosphere (a), soil (s), and estuary/water (w) as a three-node conductance network $G\psi = I$. Dissipated power $P\_TSC = \psi^{T} G \psi$ routes charge—it does not create it.

High-impedance harvesters (MEG, SMFC) benefit if a low-impedance path exists to a sink. TSC is architectural coupling for utilization, analogous to power electronics MPPT rather than a new turbine.

Conductances G_as, G_sw, G_a0 in the notebook are *illustrative Siemens* pending field instrumentation. PoC papers must label TSC figures as scenario analysis, not measured coastal data.

The column balance $P\_column = \Sigma\_k\ A\_k\ CF\_k\ \langle P''\_k \rangle$ is evaluated with lognormal uncertainty on each layer's areal power density. Replicated Monte Carlo ($N = 8000$) gives median 22.91 MW, P10 19.67 MW, P90 26.69 MW for a 1 km^2 parcel.

Layer median shares: blue_energy 1.1%, pv_hydro 98.7%, meg 0.05%, smfc 0.00%. Policy and civilizational narratives must respect this hierarchy.

```
G ψ = I

P_TSC = ψ■ G ψ

P_column = Σ_k A_k CF_k ■P''_k■
```

3.6 Layer F — PRO on anthropogenic brine

Layer F is the engineering core: PRO on anthropogenic brine. Draw concentration $c\_d \approx 1400$ mol/m^3 (~8 wt% NaCl reject) and feed $c\_f \approx 5$ mol/m^3 (treated wastewater class) give the strongest $\Delta\pi$ in our study.

$\Delta\pi = 6.916$ MPa; Kim–Baker optimum $\Delta P^* = 34.6$ bar. Water permeates according to $V\_w = L\_p\ A(\Delta\pi - \Delta P)$. Hydraulic power $P\_hyd = \rho\ V\_w\ \Delta P$; equivalent electrical power applies $\eta\_mem\ \eta\_hyd$.

SGH-1 targets $P\_target = 10$ W at $P''\_design = 8$ W/m^2, yielding $A\_mem = 0.72$ m^2 implemented as 12 plates of 200×300 mm active area each.

With default $L\_p = 1\times10^{-12}$ m/(Pa·s), steady model predicts $P = 1.66$ W ($\Pi_5 = 0.166$). Inverse sizing gives $L\_p^* = 6.03\text{e-}12$ m/(Pa·s) to hit 10 W—primary bench acceptance metric for T1 (±30%).

Concentration polarization $c\_w/c\_b = \exp(J\_w/k\_m)$ reduces effective $\Delta\pi$. Ultrasonic assist (AEH Mode B) models permeability gain g; net power subtracts P_US driver load. Feed flow Q_feed from model sets hydraulic residence time; simulated $Q = 0.149$ L/min must be validated against rotameters.

Functional requirement FR-1 mandates $0.4 \le \Delta P/\Delta\pi \le 0.6$. Numerical experiment E1 confirms maximum at 0.5. Operating outside this band sacrifices power or risks over-pressurization relative to membrane burst pressure.

$$\Delta\pi = i\ R\ T\ (c\_draw - c\_feed)$$

$$V\blacksquare\_w = L\_p\ A\ (\Delta\pi - \Delta P)$$

$$\Delta P^* = \Delta\pi\ /\ 2$$

$$P = \eta\_mem\ \eta\_hyd\ \rho\ V\blacksquare\_w\ \Delta P$$

$$c\_w\ /\ c\_b = \exp(J\_w\ /\ k\_m)$$

$$P\_net = P\_PRO(g) - P\_US - P\_0$$

3.7 UDT — Universal Differential Tink

UDT (Universal Differential Tink) is the control and actuation layer on toroidal membrane loops along a differential feed spine. A discrete **ray field** (design N_r = 64534; bench 64 transducers) delivers actuation intensity I_k at e90 wavelength λ ≈ c/f_US ≈ 5.4 cm for 28 kHz ultrasound.

Particle bytes encode geometry-scaled actuation packets: byte length ∝ (A_loop / L_line) × (λ_e90 / λ_0). The Tink kernel maps ray bytes and phases to effective mass-transfer k_m,eff, reducing concentration polarization and raising sustainable flux gain g_UDT.

Experiment E14 sweeps η_tink coupling. At default η_tink, AOR-integrated summary reports flux_gain ≈ 1.30 and f_res ≈ 694 Hz. Bench falsification: T1b compares US off vs phased on at fixed ΔP.

See docs/UDT_PHYSICS.md and simulation/differential_tink.py.

$$byte\_len \propto (A\_loop\ /\ L\_line)(\lambda\_e90\ /\ \lambda\blacksquare)$$

$$k\_m,eff = k\_m,0(1 + \eta\_tink\ w\blacksquare)$$

$$c\_w/c\_b = \exp(J\_w/k\_m,eff)$$

3.8 AOR — Acoustic-Osmotic Ram

AOR (Acoustic-Osmotic Ram) stacks three natural amplifiers: (1) resonant feed column (Q-factor) strips the CP boundary layer via acoustic streaming; (2) brine osmotic motor (Δπ MPa-class) propels permeate against hydraulic back-pressure; (3) hyperspeed converging draw leg multiplies momentum for pressure recovery via PX/turbine path.

Tagline: *Sound strips. Brine propels. Pipe amplifies.* Sound dilutes the polarization skin, not bulk brine salinity—the resource is preserved while fouling resistance improves.

E15 sweeps resonant column height H; illustrative peak P_net ≈ 3.16 W at H = 0.20 m (f_res ≈ 1875 Hz). See docs/AOR_PHYSICS.md and simulation/acoustic_osmotic_ram.py.

$$f\_res \approx c/(4H)$$

```
P_net = P_PRO(g) − P_US − P_pump + P_PX
```

3.9 VOH — Vortex-Osmotic Hydro

VOH (Vortex-Osmotic Hydro / Z-Hydro) extends AOR with rotation and vertical pressurized flux. Total pressure $P(r,z) = P_0 + \rho g z + \frac{1}{2}\rho\omega^2 r^2 + \Pi_{osm}$ combines classic z-hydro head, centrifugal rim pressure, and osmotic chemical potential—opening water-electricity sites without river dams.

Taylor–Couette shear on a halocline membrane drum strips CP; Ekman pumping at the rotating brine–fresh interface drives axial upwelling through the z-leg draw pipe.

E16 sweeps ω; flat baseline P_net = 3.16 W. Simulation breakeven ω TBD (τ_spin calibration). Milestone graph: fig16_voh_omega.png (spin vs no-spin). Physical test T1c logs P_spin_motor separately.

Civilization-scale framing: fleet deployment at every desal outfall—not sun-equivalent W/m² on acoustic panels, but GWh–TWh-class mixing entropy recovery on existing infrastructure.

See docs/VOH_PHYSICS.md and simulation/vortex_osmotic_hydro.py.

```
P = P■ + ρgz + ½ρω²r² + Π_osm

P_net,VOH = P_PRO − P_spin_motor − P_parasitics
```

4. CHORUS-Skid system architecture

CHORUS-Skid integrates SGH-1 (PRO core), AEH-1 (acoustic), CHOR-01 (enclosure), and DAQ. Design exports in exports/sgh1_design.json drive OpenSCAD via generated_constants.scad.

4.1 Functional requirements

Table 4. SGH-1 functional requirements.

ID	Requirement
FR-1	Operate at $0.4 \le \Delta P/\Delta\pi \le 0.6$
FR-2	Feed ≤ 10 bar; draw rated to ΔP^*
FR-3	Log P, Q, σ×2, T×2 continuously
FR-4	Brine leak containment (drip tray)
FR-5	Removable membrane cartridge

4.2 AEH acoustic layer

AEH-1 (Acoustic Energy Harvest) mounts on the CHORUS skid frame with six-by-four Helmholtz-inspired cells (chorus_aeh_panel.scad). **Mode A** converts incident acoustic intensity $I = p_{rms}^2/(\rho c)$ to electrical power $P = \eta I A$. At $\eta = 2\%$ and $A = 0.5\ m^2$, urban SPL 90–96 dB yields sub-watt to few-mW harvest—supplemental to DAQ, not baseload.

Mode B drives 28 kHz ultrasonic transducers on the feed manifold to disrupt CP. Modeled as multiplicative gain g on L_p, net benefit requires $P_{PRO}(g) - P_{US} - P_0 > 0$. Experiment E5 shows crossover near $g \approx 1.5$ at 1.5 W/m² ultrasonic parasitic for $A = 0.72\ m^2$.

Co-location with traffic, desal pumps, and HVAC plant noise is deliberate: Layer E in the original CHORUS vision treats cities as acoustic environments. Honest accounting keeps Mode A at mW and Mode B contingent on bench proof.

$$I = p_{rms}^2 / (\rho c)$$

$$P_{AEH} = \eta I A$$

$$P_{net} = P_{PRO}(g) - P_{US} - P_0$$

5. Numerical methods

5.1 Software modules

simulation/pro_cycle.py — steady PRO state; sizing.py — area scale law and CAD caps; membrane_transport.py — CP profile; ultrasonic_cp_gain.py — Mode B; acoustic_harvest.py — Mode A; pi_groups.py — Π_1–Π_5; experiments.py — sweeps E1–E16; differential_tink.py, acoustic_osmotic_ram.py, vortex_osmotic_hydro.py — vision stack; bench_validation.py, inverse_fit.py — T1 gates. notebooks/CHORUS_physics_proof.ipynb, CHORUS_vision_physics.ipynb, T1_bench_validation.ipynb.

5.2 Constants and assumptions

Table 5. Simulation defaults.

Parameter	Value	Source
T	298.15 K	constants.py
η_mem, η_hyd	0.35, 0.55	defaults
L_p (default)	1×10^{-12} m/(Pa·s)	placeholder until T1
ρ	1000 kg/m³	water
P''_blue anchor	15 W/m²	literature RED

5.3 Experiment matrix

Table 6. Numerical experiment matrix.

ID	Sweep	Points	Purpose
E1	ΔP/Δπ	41	Kim–Baker verification
E2	L_p	8	Inverse sizing for 10 W
E3	Salinity pairs	4	RED vs PRO Δπ
E4	J_w (CP)	6	Polarization loss
E5	Ultrasonic g	7	Mode B net gain
E6	SPL	21	Mode A harvest
E7	Column MC	8000	Layer medians
E8	c_river RED	31	P''_max vs salinity
E9	Temperature	5	Δπ(T), P(T)
E10	η_mem	11	Sensitivity
E11	Slip b	6	Nanopore G(b)

ID	Sweep	Points	Purpose
E12	Net energy	4	Parasitics + PX
E13	TSC I sweep	21	ψ, P_{diss}
E14	UDT η_{tink}	21	Flux gain, P_{net}
E15	AOR column H	6	Resonant height vs P_{net}
E16	VOH ω sweep	6	Spin vs flat breakeven

6. Results and numerical experiments

6.1 SGH-1 baseline

Table 7. SGH-1 baseline at design concentrations.

Parameter	Value
c_draw / c_feed	1400 / 5 mol/m³
$\Delta\pi$	6.916 MPa
ΔP^*	34.58 bar
A_mem	0.72 m²
P (L_p=1e-12)	1.66 W
P''	2.30 W/m²
Q_feed	0.149 L/min
E_N	144.8 mV

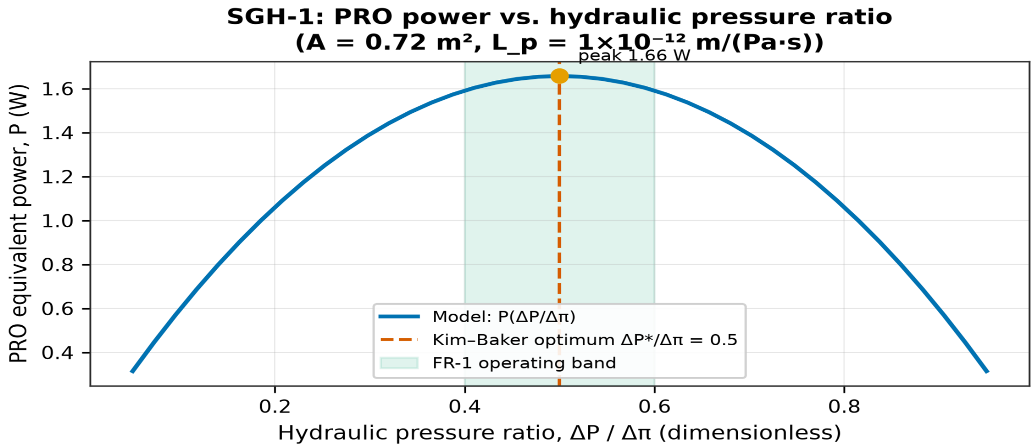
6.2 Dimensionless groups

Table 8. Dimensionless groups (pi_groups.py).

Group	Value	Meaning
Π_1	0.500	At Kim–Baker point
Π_3	0.692	Peclet order (CP)
Π_4	0.576	Area law fill
Π_5	0.166	Sim vs 10 W target
L_p*	6.03e-12	m/(Pa·s) for 10 W

6.3 Experiment E1 — Hydraulic pressure ratio

Figure 1 plots P versus $\Delta P/\Delta\pi$. Maximum at 0.5 confirms Kim–Baker. FR-1 green band 0.4–0.6 captures operable range without approaching $\Delta\pi$ (where driving force vanishes).

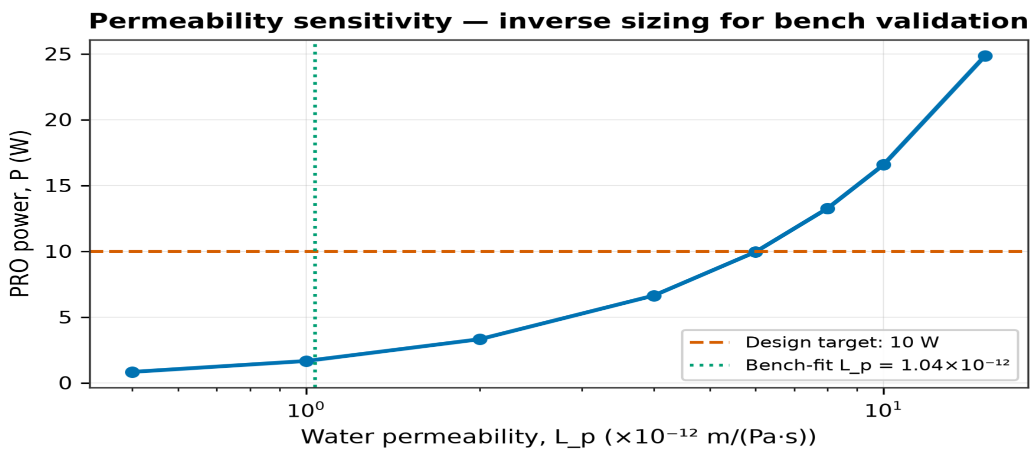


Simulated (model output): delta_P_ratio sweep, exports/paper_experiments.json

Figure 3. *PRO power versus ΔP/Δπ with FR-1 band and Kim–Baker optimum.*

6.4 Experiment E2 — Permeability

L_p* ≈ 6.03e-12 m/(Pa·s) required for 10 W. Figure 2 shows sub-target performance at literature-placeholder L_p.



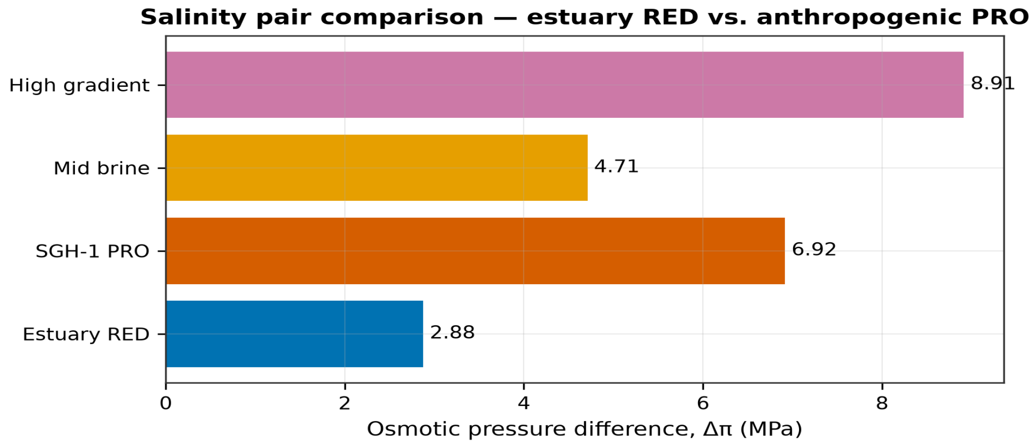
Simulated + literature-cited: L_p sweep is simulated; bench-fit marker from data/bench/T1_baseline CSV regression

Figure 4. *Power versus water permeability L_p.*

6.5 Experiment E3 — Salinity pairs

Table 9. Salinity pair comparison.

Pair	c_d	c_f	Δπ MPa	E_N mV
Estuary RED	600	20	2.88	87.4
SGH-1 PRO	1400	5	6.92	144.8
Mid brine	1000	50	4.71	77.0
High gradient	1800	2	8.91	174.8



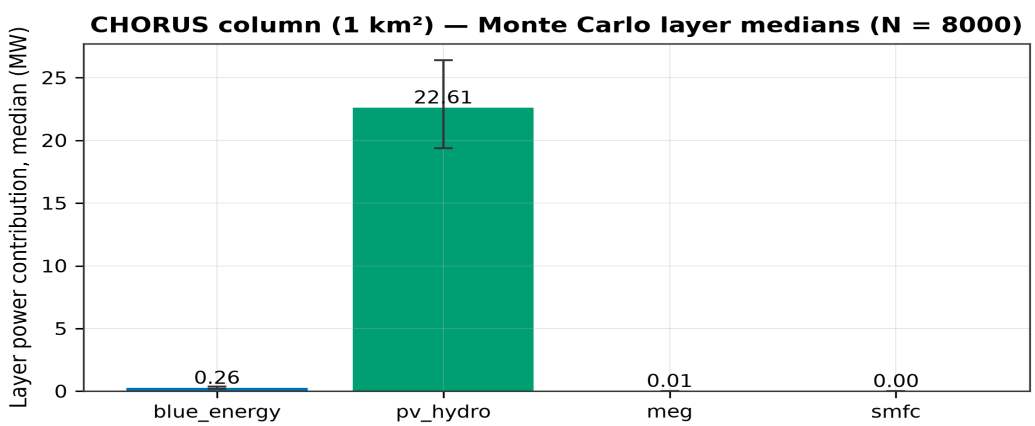
Simulated (model output): salinity_pairs, van't Hoff estimate from cited concentrations

Figure 5. *Δπ across estuary RED and anthropogenic PRO pairs.*

6.6 Experiment E7 — Column Monte Carlo

Table 10. Column layer statistics (1 km², N=8000).

Layer	Median MW	Share %	P10	P90
blue_energy	0.26	1.1	0.18	0.39
pv_hydro	22.61	98.7	19.37	26.38
meg	0.01	0.0	0.01	0.02
smfc	0.00	0.0	0.00	0.00
Total	22.91	100	19.67	26.69



Simulated (model output): column_monte_carlo layer medians, N=8000 draws with P10-P90 band

Figure 6. *Median layer contributions.*

6.7 Experiment E4 — Concentration polarization

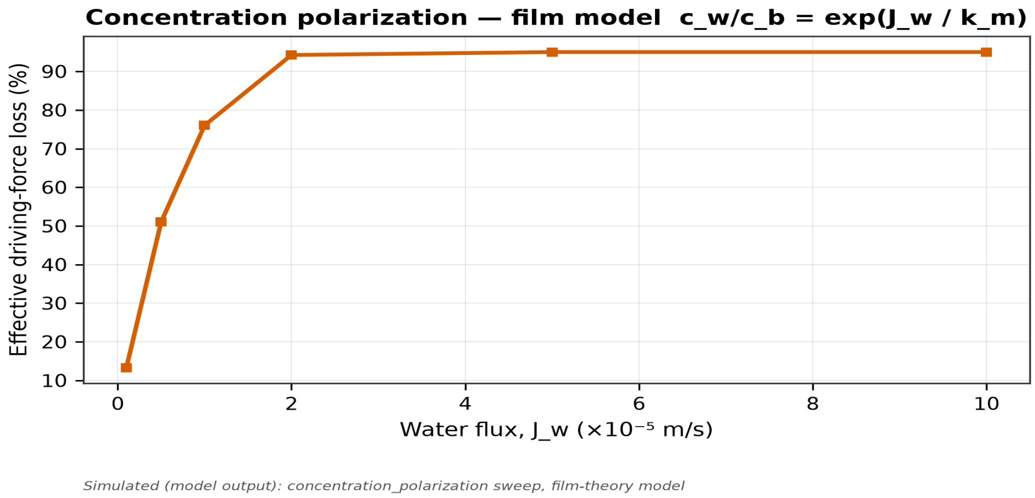


Figure 7. *Driving-force loss versus water flux.*

Table 11. CP film model results.

$J_w \times 10^{-5}$	c_w/c_b	Loss %
0.1	1.15	13.3
0.5	2.04	51.0
1.0	4.17	76.0
2.0	17.41	94.3
5.0	20.09	95.0
10.0	20.09	95.0

6.8 Experiment E5 — Ultrasonic assist

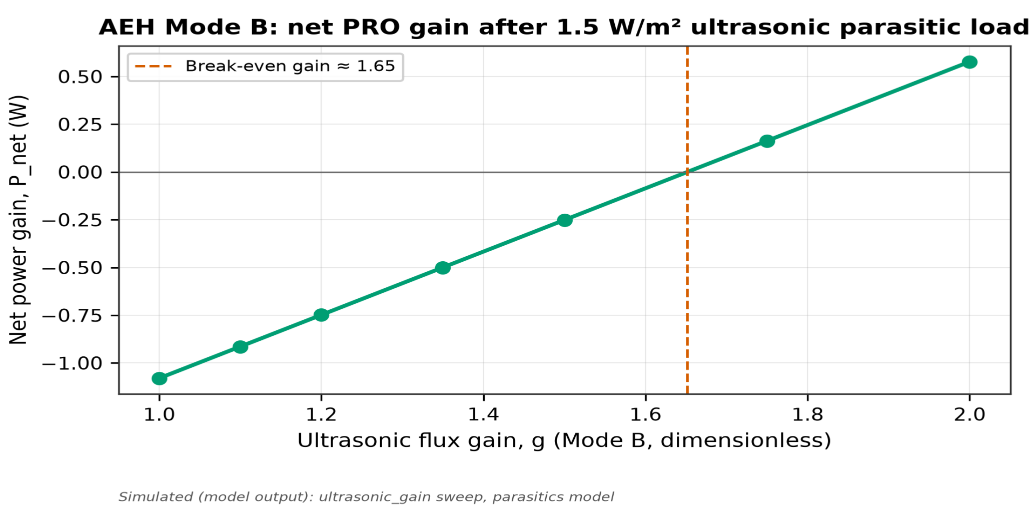
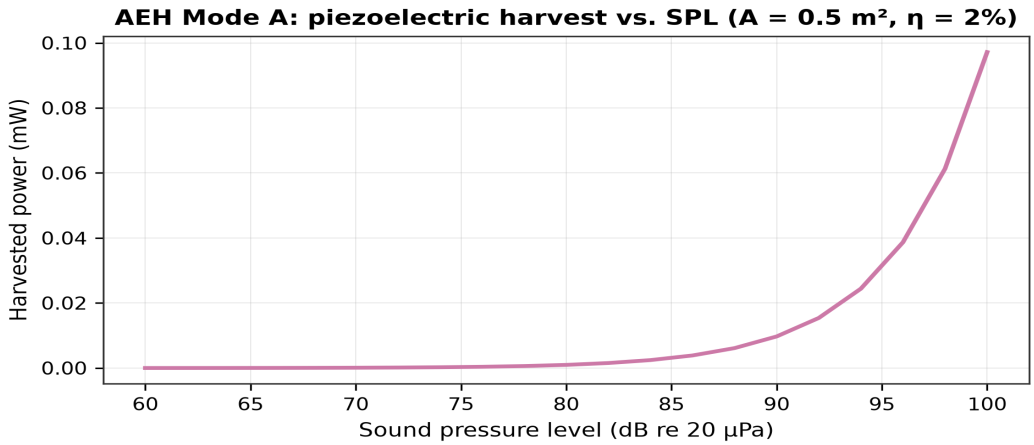


Figure 8. *Net power versus flux gain g.*

6.9 Experiment E6 — Acoustic harvest



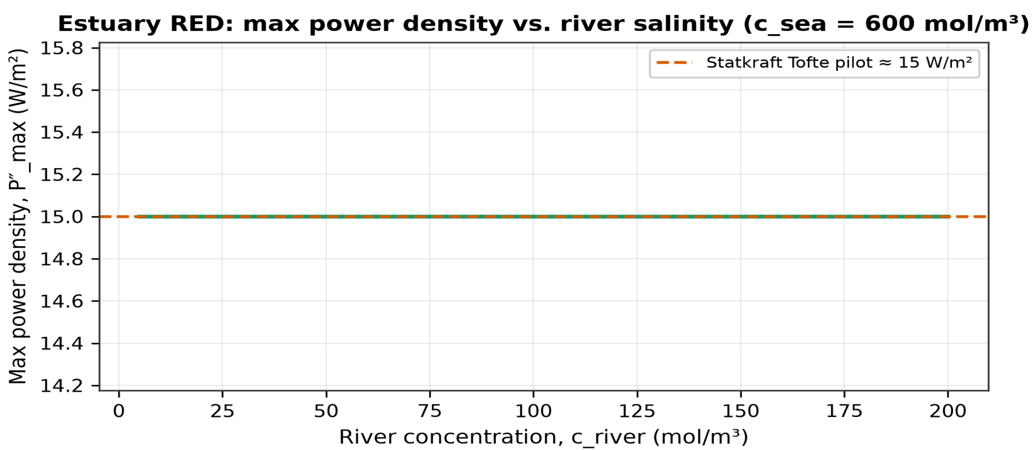
Simulated (model output): acoustic_SPL sweep, piezo transduction model

Figure 9. *Mode A harvest versus SPL.*

Table 12. Selected SPL harvest points (A=0.5 m², η=2%).

SPL dB	I W/m²	P mW
60	9.72e-07	0.000
80	9.72e-05	0.001
100	9.72e-03	0.097

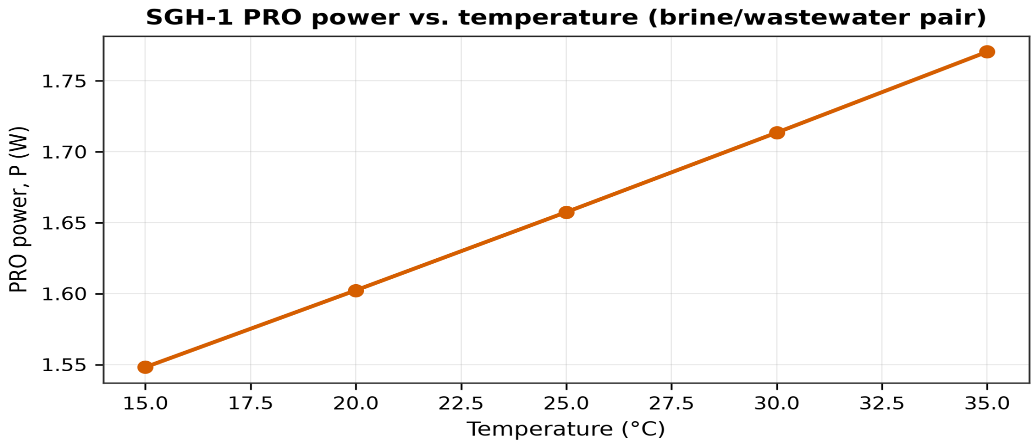
6.10 Experiment E8 — RED river salinity



Simulated + literature-cited: red_river_c sweep; reference line from exports/real_world_calibration.json

Figure 10. *Estuary RED P''_max versus c_river.*

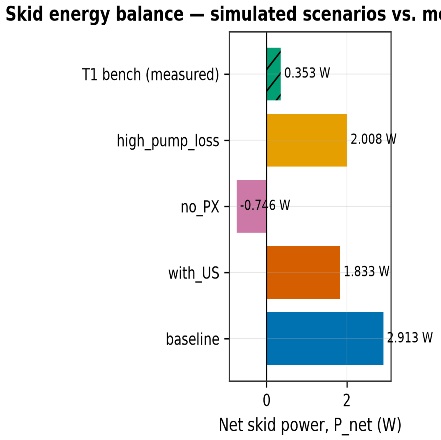
6.11 Experiment E9 — Temperature



Simulated (model output): temperature sweep, van't Hoff temperature dependence of Δπ

Figure 11. *PRO power versus temperature.*

6.12 Experiment E12 — Net skid energy



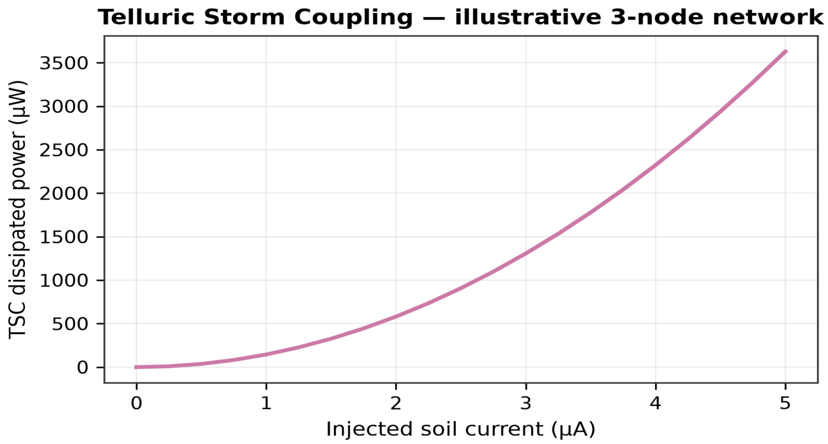
Simulated + literature-cited: simulated scenarios from parasitics model; hatched bar is bench-measured mean from data/bench/T1_baseline CSV (n=30 samples, single run)

Figure 12. *Net power scenarios with pump and PX model.*

Table 13. Skid net energy balance (simulation/parasitics.py).

Scenario	P_pro	P_pump	P_px	P_net
baseline	1.66	0.91	3.66	2.91
with_US	1.66	0.91	3.66	1.83
no_PX	1.66	0.91	0.00	-0.75
high_pump_loss	1.66	1.81	3.66	2.01

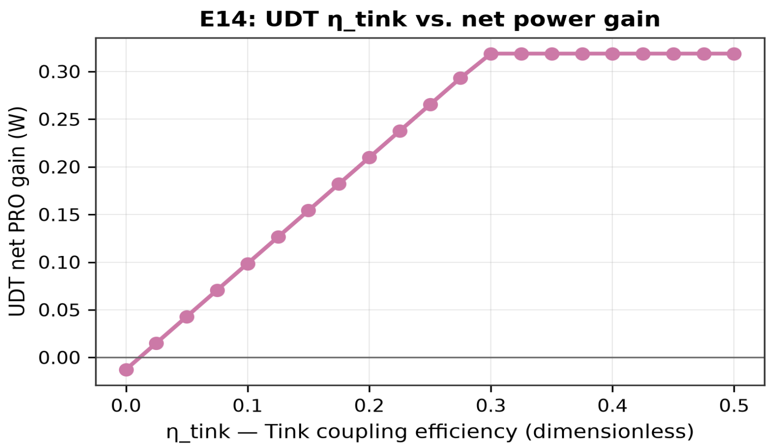
6.13 Experiment E13 — TSC injection sweep



Point estimate (no repeated-trial uncertainty data available): tsc.injection_sweep, illustrative circuit model (not bench-validated)

Figure 13. *TSC dissipated power vs soil injection current.*

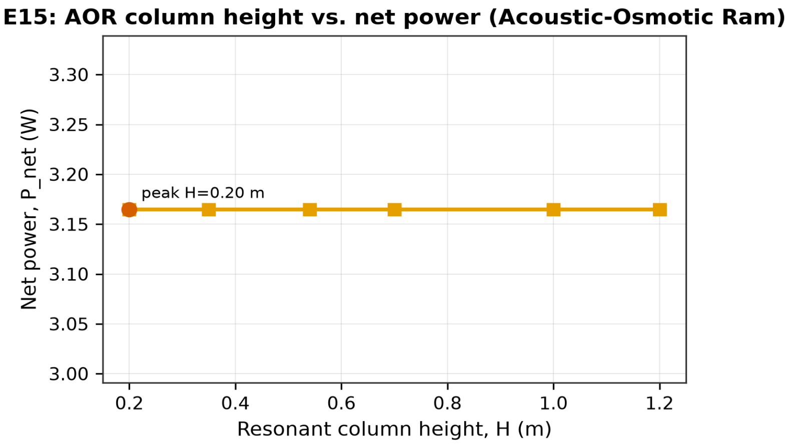
6.14 Experiment E14 — UDT η_tink sweep



Point estimate (no repeated-trial uncertainty data available): vision_stack.E14_udt_eta_tink, exploratory model (not bench-validated)

Figure 14. *UDT coupling efficiency vs net PRO gain.*

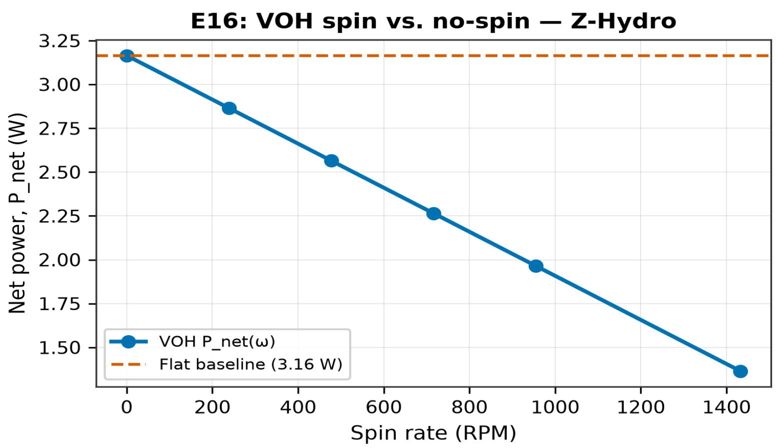
6.15 Experiment E15 — AOR resonant column



Point estimate (no repeated-trial uncertainty data available): vision_stack.E15_aor_column_height, exploratory model (not bench-validated)

Figure 15. *Column height vs net power (Acoustic-Osmotic Ram).*

6.16 Experiment E16 — VOH spin vs flat



Point estimate (no repeated-trial uncertainty data available): vision_stack.E16_voh_omega, exploratory model (not bench-validated)

Figure 16. *Z-Hydro: P_net versus spin rate with flat baseline.*

6A. Bench validation pipeline (T0–T1c)

The T0–T1 validation stack closes the loop between simulation and falsifiable bench data. daq/logger.py records test_id, phase, us_on, omega_rpm, P_net_W at ≥1 Hz. simulation/bench_validation.py inverts L_p from measured Q, ΔP, and conductivity; compares P'' to steady_state_pro prediction with ±30% T1 tolerance.

scripts/run_test_protocol.py orchestrates T0 (leak+ramp), T1 (1 h), T1b (A/B/C), and T1c (flat/spin), emitting bench_validation.json and bench_calibration.json per run.

pytest tests/ enforce math gates on pro_cycle, parasitics, vision_stack, and synthetic CSV validation. Notebooks T1_bench_validation.ipynb and VOH_spin_breakeven.ipynb are the operator-facing analysis path.

Table 14. Bench protocol gates (SGH1_TEST_PROTOCOL.md).

Stage	Duration	Pass criterion
T0	10 min leak + 30 min ramp	No leaks; positive signal 15 min
T1	1 h steady	P'' within ±30% sim; P_net > 0
T1b	3× 60 min A/B/C	P_net(C) > P_net(B) > P_net(A)
T1c	flat + spin 1 h	P_net(ω) > P_net(0) after motor subtract

7. Hardware realization

The CHORUS-Skid frame (664×480.0×900 mm) supports a membrane housing OD 280 mm, length 264 mm, bolt circle 240 mm. PRO fluid path includes feed/draw manifolds, pressure vessel rings, end caps, PX module, turbine housing, and relief valve block.

PRO-01 through PRO-09 cover the wetted stack. CHOR-01 enclosure provides thermal/moist plenum. AEH-01 panel and AEH-02 ultrasonic mount ring attach without penetrating the high-pressure draw circuit. STR-01 frame, STR-02 drip tray, STR-03 rails contain brine leaks per FR-4.

Utilities UT-01–UT-05 include pump mount, sensor bracket, and tank adapters (feed/brine/flange). v2 module RED-01 red_cartridge_v2.scad shares bolt pattern for nanopore RED experiments on the same skid.

Materials: SS316 wetted paths; EPDM gaskets; membrane sheet vendor TBD (FO/PRO class). P&ID;: low-sal tank → feed pump → prefilter → feed manifold → membrane stack → draw manifold → PX/throttle → turbine/motor load → brine return. Rupture disk at 1.2× ΔP*.

Electrical: GFCI on pump/DAQ; piezo rectifier to supplemental bus; US driver on isolated supply. See hardware/bom/SGH1_BOM.csv for 35+ line items and hardware/electrical/WIRING.md for channel map.

7.1 Safety and pressure systems

Draw-side pressure rating must exceed 1.25× test pressure per SGH1_TEST_PROTOCOL. Rupture disk set at 1.2× ΔP* ≈ 41 bar protects housing and manifolds.

Brine chemistry requires eye wash, chemical-resistant gloves, and drip tray containment (FR-4). No hydrotest with glass viewport components. Electrical GFCI on all pump and DAQ circuits.

Ultrasonic drivers are isolated from low-voltage DAQ to prevent ground loops. Acoustic panel mounting avoids resonant contact with draw high-pressure piping.

7.2 Net energy balance (parasitics)

Complete skid net power P_net = P_PRO + P_AEH + P_US,net − P_pump − P_aux + P_PX is implemented in simulation/parasitics.py and bench_validation.py. Feed pump power scales roughly with Q × ΔP_pump / η_pump. At Q = 0.15 L/min and ΔP_pump ~ 1–3 bar for pretreatment, parasitics may be comparable to modeled PRO output until PX recovery is included—PRO-07 px_module is the designed mitigation path.

Pressure exchangers can return a fraction of draw-side pressurization energy to the feed path, dramatically improving net efficiency in commercial PRO. SGH-1 includes PX housing in CAD to train integrators even before turbine/generator substitution is finalized.

DAQ budget under 2 W is negligible versus PRO target. Ultrasonic driver at 1.5 W/m² × 0.72 m² ≈ 1.08 W is material and appears in E5 as P_US.

Table 15. Representative CAD index (23 parts total).

ID	Part	Role
PRO-01	membrane_housing	Stack shell
PRO-05/06	manifolds	Feed / draw
PRO-07	px_module	Pressure exchange
AEH-01	chorus_aeh_panel	6×4 cells
CHOR-01	skid_enclosure	Plenum
STR-01	skid_frame	Structure

7A. OpenSCAD mechanical digital twin

The mechanical digital twin comprises **23 OpenSCAD parts** driven by `lib/generated_constants.scad` emitted from `simulation/run_sizing.py`. Changing P_target or plate geometry rebuilds housing OD, frame envelope, and ΔP_bar in one command—eliminating unit drift between simulation, drawings, and STL export.

PRO housing: cylinder OD 280 mm, stack length 184 mm (n_plates=12 × pitch 12 mm + 40 mm end allowance), bore ID ≈ OD − 20 mm wall. Active window per plate 200×300 mm → A_mem = 0.72 m² total.

AEH panel: 24 Helmholtz cells (288×400×28 mm), neck Ø8 mm, cavity Ø35 mm. Piezo pockets on rear face; 28 kHz US boss on feed-side assist path.

Assemblies `sgh1_assembly.scad` and `sgh1_exploded_assembly.scad` document install order. v2 `sgh1_red_cartridge_v2.scad` shares bolt circle for RED nanopore experiments. See docs/hardware/OPENSCAD_DEEP_DIVE.md and exports/openscad_audit.json.

Table 16. Generated constants driving all PRO parts.

CAD parameter	Value
housing_od	280 mm
stack_length	184 mm
frame_L×W×H	664×480×900 mm
bolt_circle	240 mm
delta_P_bar	34.6 bar

Table 17. OpenSCAD audit (12 of 23 parts; full list in openscad_audit.json).

Part file	Modules	Sized from JSON
chorus_aeh_panel.scad	helmholtz_cell, aeh_panel	yes
chorus_skid_enclosure.scad	chorus_enclosure, chorus_enclosure_part	yes
sgh1_assembly.scad	—	yes
sgh1_brine_tank_adapter.scad	brine_tank_adapter	no
sgh1_drip_tray.scad	drip_tray	yes
sgh1_end_cap.scad	end_cap	yes
sgh1_exploded_assembly.scad	—	yes
sgh1_feed_tank_adapter.scad	feed_tank_adapter	no
sgh1_flange_adapter.scad	flange_adapter	no
sgh1_manifold_draw.scad	draw_manifold, sgh1_manifold_draw_part	yes

Part file	Modules	Sized from JSON
sgh1_manifold_feed.scad	feed_manifold, sgh1_manifold_feed_part	yes
sgh1_membrane_housing.scad	housing_shell, sgh1_membrane_housing_part	yes

8. Bench and field test protocol

T0 — Coupon test: Air leak test 3 bar, 10 min; water flush; establish feed 5 g/L and draw 80 g/L simulant; ramp ΔP from $0 \rightarrow 0.5\Delta\pi \rightarrow \Delta P^*$ over 30 min; log at 1 Hz. Pass: no leaks, positive steady power 15 min.

T1 — Full stack: Hydrotest draw at 1.25× ΔP^*; 1 h steady run; compare P'' to simulation ±30%. This directly tests L_p calibration and CP losses.

T2 — Field sidestream: Utility NDA; tie-in to real brine and effluent; replicate logging.

AEH tests: SPL meter at panel; Voc into 1 MΩ; US on/off comparison for Mode B flux gain.

T1b — UDT/AOR: Runs A (US off), B (phased UDT), C (full AOR resonant column); pass if P_net(C) > P_net(B) > P_net(A) over 60 min at fixed ΔP.

T1c — VOH/Z-Hydro: Flat (ω=0) vs spinning drum; log P_spin_motor separately; pass if P_net(ω) > P_net(0) after spin parasitic subtraction.

Automated validation: simulation/bench_validation.py applies ±30% T1 gate on exported CSV; scripts/run_test_protocol.py orchestrates full protocol runs.

8A. Worked numerical examples

Worked example 1 (estuary RED): With $c_{sea} = 600$, $c_{river} = 20$ mol/m³, $T = 298.15$ K, $E_N = 87.39$ mV per pair. Fifty pairs → $V_{oc} = 4.37$ V. If $P''_{max} = 15$ W/m², then $R_{int} = V_{oc}^2/(4 P'') = 0.318$ Ω·m².

Worked example 2 (brine PRO): $c_{draw} = 1400$, $c_{feed} = 5$ → $\Delta\pi = 6.92$ MPa. At $\Delta P = \Delta\pi/2$, $L_p = 1\times10^{-12}$, $A = 0.72$ m², $\eta_{mem}\, \eta_{hyd} = 0.35\times0.55$ → $P \approx 1.66$ W. To reach 10 W, scale L_p to 6.03e-12 m/(Pa·s) or add membrane area beyond twelve plates.

Worked example 3 (mixing ceiling): $\Delta\pi_{estuary} \times Q = 1438$ MW at $Q = 500$ m³/s illustrates that bulk mixing power is enormous but inaccessible without controlled interface area and membrane technology.

8B. Ranked concepts and deployment path

The CHORUS notebook exports four ranked concept tags used throughout the program:

Safest: PRO salinity-gradient core. **Smartest:** Ultrasonic CP gain + PRO. **Strangest:** TSC + AEH urban panel. **Civilization-scale:** Co-located desal brine + effluent + CHORUS skid.

These tags are communication devices, not performance guarantees. Safest maps to Tier-1 bench PRO. Smartest maps to AEH Mode B contingent on positive E5 net gain. Strangest maps to Tier-2 TSC. Civilization maps to column Monte Carlo only.

Deployment targets co-located desalination and wastewater treatment plants where brine and effluent are available without new intake infrastructure. SGH1_PATENT_AND_DEPLOYMENT.md outlines claim focus on PRO-on-brine plus ultrasonic CP assist plus integrated DAQ—not on column MW totals.

Outreach sequence: (1) bench T0/T1 at lab simulant; (2) pilot T2 at utility sidestream; (3) scale membrane area or plate count toward kW-class if L_p validates. Regulatory path follows existing industrial water handling, not new estuary impingement.

Economic viability requires net positive skid power after pump, pretreatment, and US parasitics—a calculation deferred until parasitics.py is integrated. Even at 10 W bench proof, the learning value is membrane characterization under real Δπ, not immediate LCOE competitiveness.

8C. Notebook reproducibility map

notebooks/CHORUS_physics_proof.ipynb implements §I–VIII: symbolic Gibbs and Nernst (SymPy); RED slip conductance sweep (NumPy); MEG flux (NumPy); PV thermal ODE (SciPy solve_ivp); Butler–Volmer SMFC; global circuit; TSC Kirchhoff; column Monte Carlo with boxplots.

The final cell exports chorus_results.json consumed by this paper and pdf-genesis. notebooks/SGH1_PRO_simulation.ipynb repeats PRO sizing with AEH and vision-stack sweeps. notebooks/T1_bench_validation.ipynb ingests bench CSV and runs ±30% T1 gate. notebooks/VOH_spin_breakeven.ipynb produces the spin-vs-flat milestone graph. notebooks/CHORUS_vision_physics.ipynb derives UDT/AOR/VOH equations with inline NumPy.

Re-running notebooks after parameter changes is mandatory for publication integrity—JSON exports are the single source of truth for tables in Section 6 and Appendix A.

9. Discussion

Sidestream PRO wins the near-term race because infrastructure exists: tanks, pumps, permits, and operators. Estuary RED requires greenfield environmental permitting and long membrane interfaces in biofouling waters.

The 1.66 W vs 10 W gap is the central honest tension. It is not fraud—it is an explicit call for membrane characterization. Literature PRO foils after conditioning often exceed bare 1×10^{-12} m/(Pa·s).

Column median 22.9 MW/km² is a thought experiment with PV-dominated area weights. Investors and policymakers must not conflate it with SGH-1 bench watts.

TSC and global circuit layers are valuable for systems storytelling and patent differentiation but must not appear in spec sheets as measured performance.

Ultrasonic CP assist is promising where feed salinity is high and J_w is large enough that polarization loss exceeds driver parasitics—exactly the brine/effluent regime.

Open-source release (differential-harness) enables third-party falsification—critical for pre-revenue hardware science.

9.1 Claims hierarchy (summary)

Table 18. Claims tiering for stakeholders.

Tier	Claim	Evidence status
1	PRO $\Delta\pi$, ΔP^* sizing	Model + CAD
1	Kim–Baker peak	E1 sweep
1	L_p calibration needed	E2, Π_5
2	Column 22 MW/km²	E7 MC, PV-dominated
2	TSC routing	Illustrative G
2	AEH mW harvest	E6, Mode A

10. Conclusion and future work

CHORUS-SGH-1 delivers a complete, falsifiable pipeline from osmotic thermodynamics to bench hardware and a three-layer vision stack (UDT, AOR, VOH) aimed at osmotic-vortex hydro at desal outfalls. Black & White PoC report v5 adds layer theory A–F, vision physics 3.7–3.9, experiments E1–E16, sixteen figures, T0–T1c validation code, and explicit tiering of claims. The central engineering action is bench validation of L_p and CP on anthropogenic brine—then T1b/T1c vision tests—before utility sidestream outreach.

10.1 Future work

Physical T0/T1 on assembled stack; publish bench CSV with inverse L_p fit (inverse_fit.py).

T1b/T1c vision validation: UDT phased rays and VOH spin breakeven on real brine.

CAD v3: sgh1_tink_ring, sgh1_z_pipe, sgh1_vortex_basin for toroidal loop and z-leg harvest.

Field T2 at co-located desal + WWTP under utility NDA.

Couple salt permeability B to draw dilution over time (transient PRO).

Fleet-scale modeling: N modules at global brine outfall discharge (~142×10^6 m^3/day).

Peer-reviewed submission with verified DOIs; Connor White co-authorship on vision experiments.

Appendix A — Estuary RED reference values

Table 19. From chorus_results.json.

Quantity	Value
E_N	87.39 mV
V_stack (50 pair)	4.37 V
Δπ	2.876 MPa
P''_blue	15 W/m²
R_int	0.318 Ω·m²
P_mix ceiling	1438 MW

Appendix B — E1 pressure-ratio sweep (complete, 41 points)

Table 20. Full Kim–Baker sweep (experiment E1).

ΔP/Δπ	ΔP MPa	P (W)	P'' (W/m²)	Q (L/min)
0.050	0.346	0.315	0.437	0.284
0.073	0.501	0.446	0.619	0.277
0.095	0.657	0.570	0.792	0.270
0.118	0.813	0.687	0.955	0.264
0.140	0.968	0.798	1.109	0.257
0.162	1.124	0.902	1.253	0.250
0.185	1.280	1.000	1.388	0.244
0.208	1.435	1.090	1.514	0.237
0.230	1.591	1.174	1.631	0.230
0.253	1.746	1.251	1.738	0.223
0.275	1.902	1.322	1.836	0.217
0.297	2.058	1.386	1.924	0.210
0.320	2.213	1.443	2.004	0.203
0.342	2.369	1.493	2.074	0.196
0.365	2.524	1.537	2.134	0.190
0.387	2.680	1.574	2.186	0.183

ΔP/Δπ	ΔP MPa	P (W)	P'' (W/m²)	Q (L/min)
0.410	2.836	1.604	2.227	0.176
0.432	2.991	1.627	2.260	0.170
0.455	3.147	1.644	2.283	0.163
0.477	3.303	1.654	2.297	0.156
0.500	3.458	1.657	2.302	0.149
0.522	3.614	1.654	2.297	0.143
0.545	3.769	1.644	2.283	0.136
0.568	3.925	1.627	2.260	0.129
0.590	4.081	1.604	2.227	0.123
0.613	4.236	1.574	2.186	0.116
0.635	4.392	1.537	2.134	0.109
0.657	4.547	1.493	2.074	0.102
0.680	4.703	1.443	2.004	0.096
0.703	4.859	1.386	1.924	0.089
0.725	5.014	1.322	1.836	0.082
0.748	5.170	1.251	1.738	0.075
0.770	5.326	1.174	1.631	0.069
0.792	5.481	1.090	1.514	0.062
0.815	5.637	1.000	1.388	0.055
0.838	5.792	0.902	1.253	0.049
0.860	5.948	0.798	1.109	0.042
0.883	6.104	0.687	0.955	0.035
0.905	6.259	0.570	0.792	0.028
0.927	6.415	0.446	0.619	0.022
0.950	6.570	0.315	0.437	0.015

Appendix B2 — SymPy symbolic verification

Independent symbolic algebra (simulation/symbolic_checks.py) reproduces key limits:

Brine pair $\Delta\pi$ = 6.916 MPa (numeric subs), E_N = 144.8 mV.

Differentiating P_hyd ∝ (Δπ−ΔP)ΔP gives critical point dpi/2 — confirming Kim–Baker without discrete sweep.

LaTeX forms are exported to exports/symbolic_checks.json for thesis appendices.

```
R T i \left(c_{h} - c_{l}\right)

Numeric Δπ = 6.9159 MPa

Kim–Baker critical: dpi/2
```

Appendix B3 — Draft patent claims

Draft independent claims (not filed — not legal advice):

(1) An integrated energy skid comprising a pressure-retarded osmosis membrane stack fed by anthropogenic desalination reject brine as draw and treated wastewater as feed; a moisture/thermal plenum enclosure; an acoustic panel with piezoelectric harvest and ultrasonic transducer on the feed path; and a shared data-acquisition bus correlating pressure, flow, and conductivity.

(2) The skid of claim 1 wherein hydraulic pressure on the draw side is controlled to 0.4–0.6 of osmotic pressure difference via feedback from dual conductivity sensors.

(3) A method of enhancing PRO permeate flux by applying ultrasonic energy to reduce concentration polarization while subtracting transducer parasitic load from net power accounting.

(4) A Telluric Storm Coupling conductance network routing charge among atmosphere, soil, and water nodes to improve utilization of high-impedance moist-electric or microbial harvesters.

(5) UDT: geometry-scaled particle bytes on a discrete ray field controlling k_m,eff on toroidal membrane loops.

(6) AOR: resonant acoustic column coupled to brine osmotic motor and hyperspeed draw ram leg.

(7) VOH: rotating halocline cell with axial z-leg pressurized flux harvest (Z-Hydro).

Prior art search targets: Statkraft PRO, SWRO–PRO hybrids, Taylor–Couette membrane shear, phased ultrasound CP literature, and desal sidestream valorization patents.

Appendix C — E2 L_p sweep (full)

$L_p \times 10^{-12}$	P (W)	P''	≥10W
0.5	0.83	1.15	N
1.0	1.66	2.30	N
2.0	3.31	4.60	N
4.0	6.63	9.21	N
6.0	9.94	13.81	N
8.0	13.26	18.42	Y
10.0	16.57	23.02	Y
15.0	24.86	34.53	Y

Appendix D — E5 ultrasonic (full)

g	P_base	P_us	P_net
1.00	1.66	1.08	-1.080
1.10	1.66	1.08	-0.914
1.20	1.66	1.08	-0.749
1.35	1.66	1.08	-0.500
1.50	1.66	1.08	-0.251
1.75	1.66	1.08	0.163
2.00	1.66	1.08	0.577

Appendix E — Bill of materials (excerpt)

Table 23. SGH1_BOM.csv (first 20 lines; full BOM in repository).

ID	Name	Qty	Material
FRAME-001	8020 extrusion 40x40mm	12m	6061-T6 aluminum
FRAME-002	Skid foot pad	4	rubber/mount
HOUS-001	Membrane housing shell	1	SS316L or HDPE
HOUS-002	End cap feed	1	SS316L
HOUS-003	End cap draw	1	SS316L

ID	Name	Qty	Material
HOUS-004	Pressure ring	2	SS316L
MEM-001	Membrane plate/spacer	1	HDPE+CNC
MEM-002	Commercial PRO/FO membrane s	1	composite OEM
MEM-003	Viton O-ring housing	2	rubber
MANF-001	Feed manifold	1	PP or SS316
MAND-001	Draw manifold rated pressure	1	SS316
PUMP-001	Low-sal feed pump	1	12V diaphragm
PUMP-002	Prefilter 10um	1	polypropylene
PX-001	Pressure exchanger or needle	1	SS316
SENS-001	Pressure transducer 0-40 bar	2	piezo
SENS-002	Conductivity probe	2	SS
SENS-003	Flow meter	2	magnetic or turbin
DAQ-001	Raspberry Pi 4 + ADC HAT	1	electronics
CHOR-001	Thermal plenum enclosure	1	HDPE/PP
AEH-001	Acoustic panel with Helmholt	1	PLA+PVDF

Appendix F — CAD file list (complete)

- hardware/openscad/sgh1_assembly.scad
- hardware/openscad/sgh1_exploded_assembly.scad
- hardware/openscad/sgh1_membrane_housing.scad
- hardware/openscad/sgh1_membrane_plate.scad
- hardware/openscad/sgh1_manifold_feed.scad
- hardware/openscad/sgh1_manifold_draw.scad
- hardware/openscad/sgh1_px_module.scad
- hardware/openscad/sgh1_turbine_housing.scad
- hardware/openscad/chorus_aeh_panel.scad
- hardware/openscad/chorus_skid_enclosure.scad
- hardware/openscad/sgh1_skid_frame.scad
- hardware/openscad/sgh1_red_cartridge_v2.scad

Appendix G — Real-world calibration JSON

Perth sidestream: $\Delta\pi$=5.92 MPa, P=1.22 W. Mixing energy seawater+brine: 0.14 kWh/m³ (lit.) vs model 0.826 kWh/m³.

Appendix H — Reproducibility commands

```
python -m simulation.experiments

python -m simulation.pi_groups

python -m simulation.symbolic_checks

python scripts/audit_openscad.py

python simulation/run_sizing.py --power 10 --density 8

python scripts/generate_paper_figures.py

python scripts/build_research_paper.py

pytest
```

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